

Quick Reference for the Gleacher Game

Factory

Cobb-Douglas Production Function (yearly throughput): $Y = AK^{\alpha}L^{\beta}$

Cobb-Douglas Production Function (daily throughput): $\lambda = \frac{AK^{\alpha}(l*364)^{\beta}}{364}$

Little's Law: $INV = \lambda * CT$

Bench set up time: 0.05 days; **A:** 0.009; **α :** 0.10; **β :** 0.85

Line set up time: 0.5 day; **A:** 0.010; **α :** 0.30; **β :** 0.75

Cell set up time: 1 day; **A:** 0.020; **α :** 0.80; **β :** 0.30

Starting factory type: Bench

Starting factory investment: \$100,000 capital + \$100,000 land

Starting daily expenditure: \$2,500

Minimum initial investment in line factory: \$500k capital + \$100k land

Minimum initial investment in cell factory: \$3m capital + \$100k land

Initial Construction Time: 90 days

Add-on Capex: 30 days

Capital Depreciation: 15 years

Bass Model

Formula for daily innovator arrivals: $p * \text{remaining market size}$

Formula for daily arrivals from advertising: $p * \text{remaining market size} * (\$ \text{adv} / \$500)$

Formula for daily imitator arrivals: $q * \text{remaining market size} * (\text{cumulative sales} / \text{remaining mkt size})$

Minimum order inter-arrival time (for a given product in a particular region): 10 seconds

Maximum customer arrival rate (for a given product in a particular region): 8640 units per day

Distribution Center

Investment: \$2,500,000 capital + \$100,000 land

Construction Time: 60 days

Capital Depreciation: 15 years

Daily Expenditure: \$2,000

Handling for retail sales: \$10/unit

Commission for retail sales: 20% of revenue

Shipping from Factory to DC *(paid by the shipper)*

Container in region: \$5,000 per 1,000 units; 7 days

Container between region: \$10,000 per 1,000 units; 21 days

Mail in region: \$200 per 10 units; 1 day

Mail between regions: \$400 per 10 units; 3 days

Shipping from DC to DC (*paid by the shipper*)

Container in region: \$5,000 per 1,000 units; 7 days

Container between region: \$10,000 per 1,000 units; 21 days

Mail in region: \$200 per 10 units; 1 day

Mail between regions: \$400 per 10 units; 3 days

Focus Group

Size: 10 participants

Cost: \$20,000

Time: 7 days

Financial

Asset cost of capital: 15% APR

Excellent debt: 10% APR; 20x interest coverage ratio

Good debt: 15% APR; 7x interest coverage ratio

Poor debt: 25% APR; 2x interest coverage ratio

Emergency Loan: 40% APR

Interest earned on cash balances: 3% APR

Dividends: 6.5% APR after-tax return on any dividends that are paid out to shareholders

Return to Investors: cash on hand - debt + dividends paid + after tax return on dividends distributed

Trade credit terms for raw materials payable: 30 days to pay

Trade credit terms for other payables: 15 days to pay

Tax rate: 35%

Taxes Payable accumulate daily, and the account balance is paid on the last day of each quarter